TSM 099 Quadratic and Rational Equations Test

Practice 3 | Version 202610122020.27

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right.

QUESTIONS	ANSWERS
1. Solve. List excluded values and check candidates in the original equation. $\frac{8}{x^2-16}+\frac{x-5}{x-4}=\frac{9}{x+4}$	1. Excluded values 4 pts Candidates/check 6 pts Solution 10 pts
2. Solve. List excluded values and check in the original equation. $\frac{8}{2x-6}+\frac{13}{x^2-9x+18}=\frac{5}{x-6}$	2. Excluded values 5 pts Solution/check 15 pts
3. Solve with the quadratic formula. Simplify radicals. $2x^2+3x-4=0$	3. Solutions 20 pts
4. Solve by factoring. $x^2+3x-18=0$	4. Solutions 20 pts
5. Solve by taking square roots; do not expand the square. $2(x-1)^2=32$	5. Solutions 20 pts

Instructor answer key: TSM 099 Quadratic and Rational Equations Test

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Each column matches the vertical answer positions on its corresponding test page.

Concise grading reference; required work and checks are assessed on separate paper.

TEST PAGE 1

1. Excluded values

$\{-4, 4\}$ — 4 pts

Candidates/check

$x = 4$ (rejected), $x = 6$ — 6 pts

Solution

$x = 6$ — 10 pts

2. Excluded values

$\{3, 6\}$ — 5 pts

Solution/check

$x = 4$ — 15 pts

3. Solutions

$x = -\frac{3}{4} + \frac{\sqrt{41}}{4}, \quad -\frac{\sqrt{41}}{4} - \frac{3}{4}$ — 20 pts

4. Solutions

$x = -6, \ 3$ — 20 pts

5. Solutions

$x = -3, \ 5$ — 20 pts